

Advance Analysis of Algorithms

Graph Algorithms

Breadth First Search (BFS)

Depth First Search (DFS)

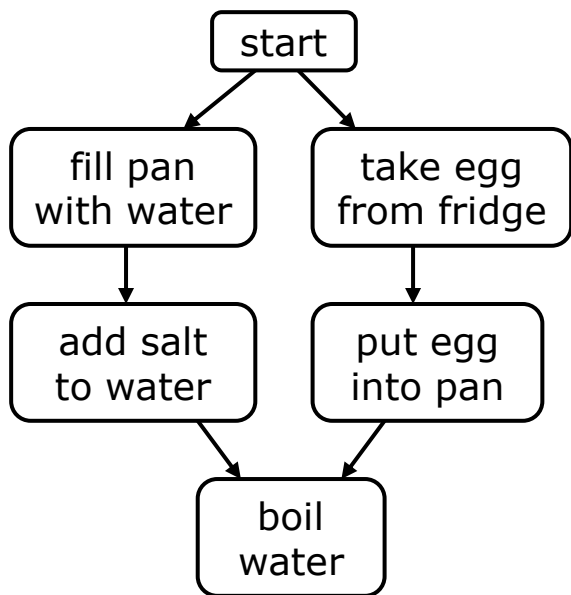
Zohaib Ali

Department of Computer Science

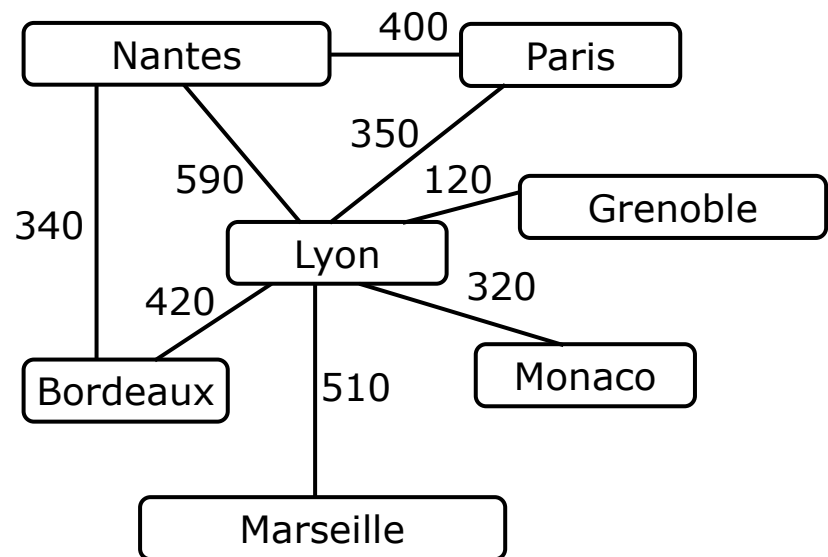
Graph Fundamentals

- A graph is a collection of *nodes* (or vertices, singular is vertex) and *edges* (or arcs)
 - » Each *node represents an element*
 - » Each edge connects *two nodes* together (or possibly the same node to itself) and may contain an *edge attribute*
- There are two kinds of graphs: *directed graphs* (sometimes called digraphs) and *undirected graphs*
 - » A directed graph is one in which the edges have a *direction*
 - » An undirected graph is one in which the edges *do not have a direction*.

Graph Fundamentals



A directed graph



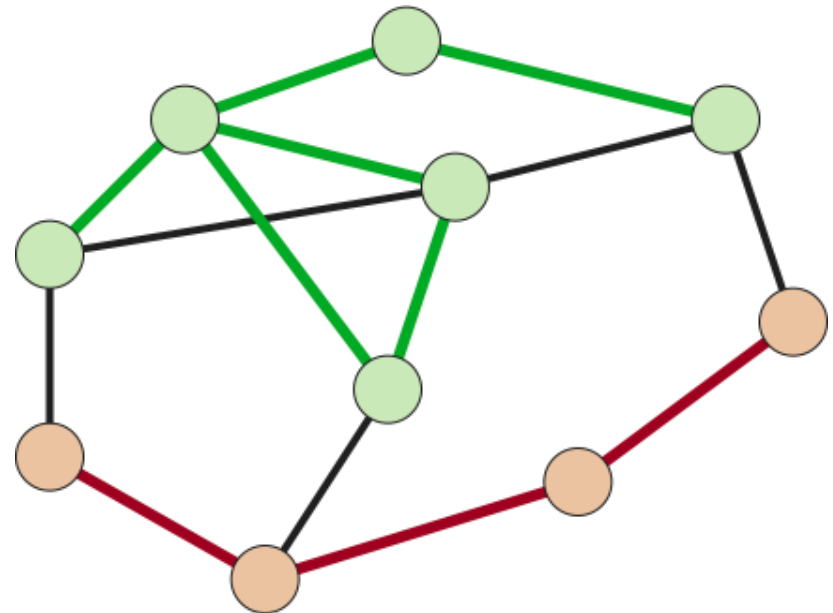
An undirected graph

Graph Fundamentals

- If two nodes are *connected by an edge*, they are neighbors (and the nodes are adjacent to each other)
- The *degree of a node* is the number of edges it has
- For directed graphs,
 - » If a *directed edge* goes from node *S to node D*, we call *S the source* and *D the destination* of the edge
 - The edge is *an out-edge of S* and an *in-edge of D*
 - S is a predecessor of D, and D is a successor of S
 - » The *in-degree* of a node is the number of in-edges it has
 - » The *out-degree* of a node is the number of out-edges it has

Graph Fundamentals

- A **path** is a sequence of vertices $P = (v_0, v_1, \dots, v_k)$ such that, for $1 \leq i \leq k$, edge $(v_{i-1}, v_i) \in E$.
- Path P is **simple** if no vertex appears more than once in P .
- A cycle is a path whose **first and last** nodes are the **same**



$P_1 = (g, d, e, b, d, a, h)$ is not simple.

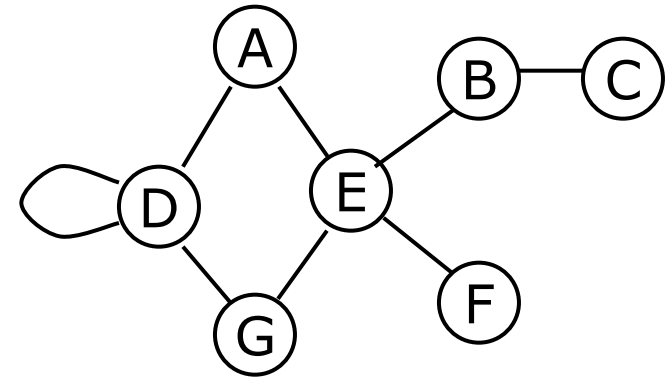
$P_2 = (f, i, c, j)$ is simple.

Graph Representation

- Adjacency-matrix Representation
- Adjacency Lists Representation

Adjacency-matrix Representation

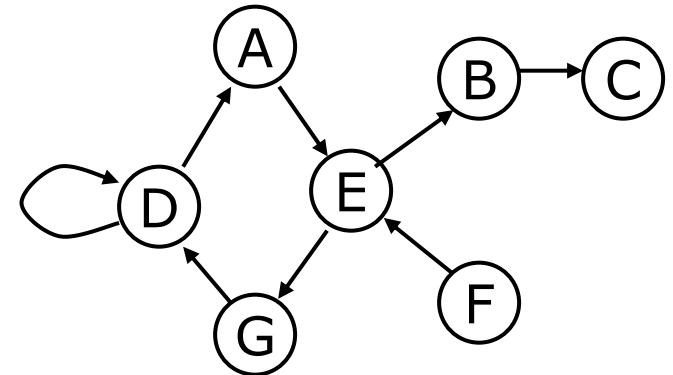
- A 2-D array has a mark at $[i][j]$ if there is an edge from node i to node j
- The adjacency matrix is symmetric about the main diagonal



	A	B	C	D	E	F	G
A				●	●		
B			●		●		
C		●					
D	●			●			●
E	●	●				●	●
F					●		
G				●	●		

Adjacency-matrix Representation

- An adjacency matrix can equally well be used for *directed graphs*.
- A 2-D array has a mark at $[i][j]$ if there is an edge from node i to node j



	A	B	C	D	E	F	G
A					●		
B			●				
C							
D	●			●			
E		●					●
F					●		
G				●			

Adjacency-Matrix Representation

- Pros:
 - » Simple to implement
 - » Easy and fast to tell if a pair (i,j) is an edge: simply check if $A[i][j]$ is 1 or 0
- Cons:
 - » No matter how few edges the graph has, the matrix takes $O(n^2)$ in memory

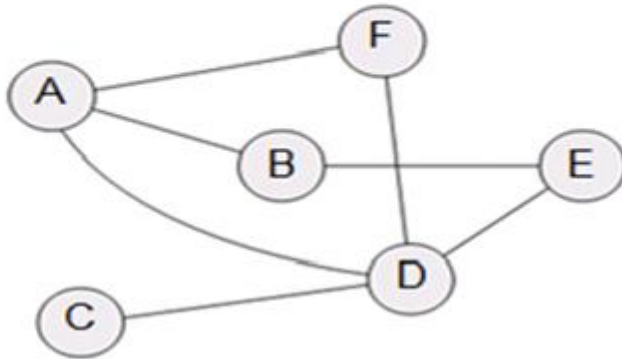
Adjacency-list Representation

- A graph of n nodes is represented by a *one-dimensional array L of linked lists*, where
 - » $L[i]$ is the linked list containing all the nodes *adjacent to node i*
 - » The nodes in the list $L[i]$ are in *no particular order*

Adjacency-list Representation

Graphs and Digraphs — Examples

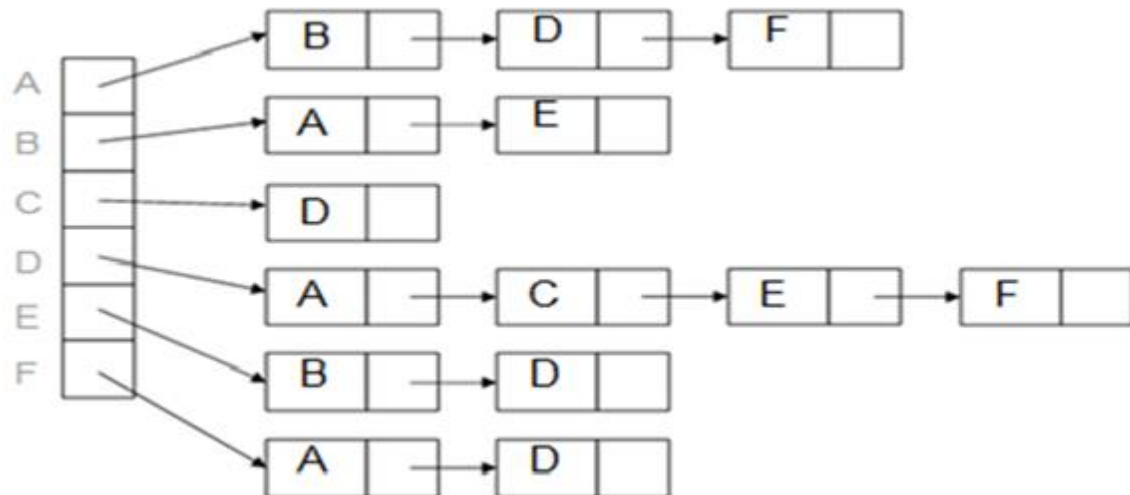
An (undirected) graph $G = (V, E)$



adjacency matrix for G

	A	B	C	D	E	F
A	0	1	0	1	0	1
B	1	0	0	0	1	0
C	0	0	0	1	0	0
D	1	0	1	0	1	1
E	0	1	0	1	0	0
F	1	0	0	1	0	0

adjacency list for G



Adjacency-list Representation

- Pros:
 - » *Saves on space* (memory): the representation takes as many memory words as there are nodes and edge.
- Cons:
 - » It can take up to *$O(n)$ time* to determine if a *pair of nodes* (i,j) is an edge: one would have to search the linked list $L[i]$, which takes time proportional to the length of $L[i]$.

Graph Traversal

- *Graph traversal* is the process of writing out all the nodes of a simple, connected graph G in some *organized way*.
- Algorithms, that apply for generalize traversal of any graph. They are:
 - » Breadth First Search (Queue)
 - » Depth First Search (Stack)

Graph Traversal: BFS

- The state of a vertex, u , is stored in a color variable as follows:
 - » $\text{color}[u] = \text{White}$ - for the "*undiscovered*" state,
 - » $\text{color}[u] = \text{Green}$ - for the "*discovered but not fully explored*" state,
 - » $\text{color}[u] = \text{Black}$ - for the "*fully explored*" state.

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

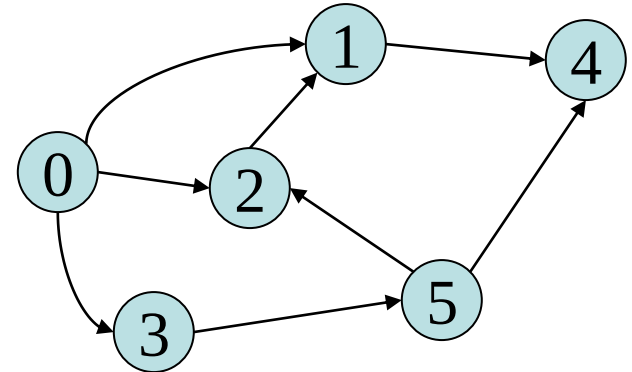
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex
 $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

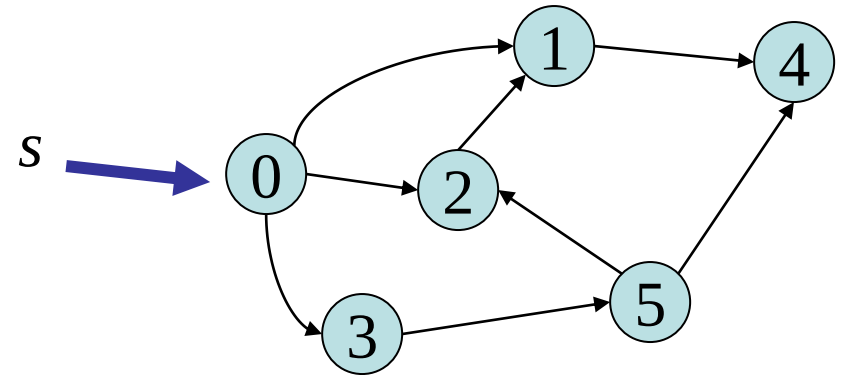
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q = \emptyset$

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

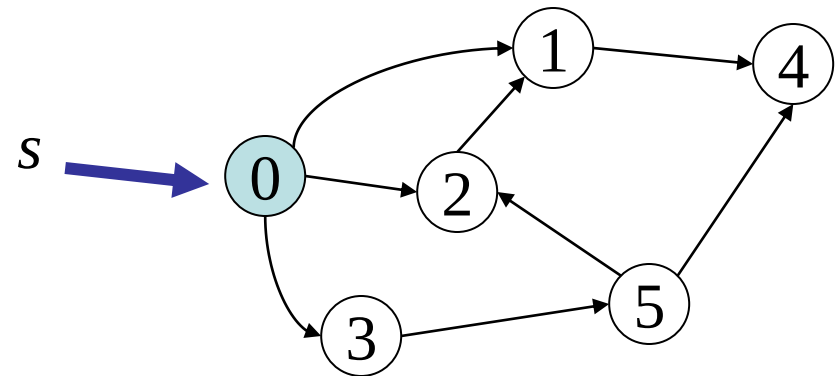
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q = \emptyset$

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

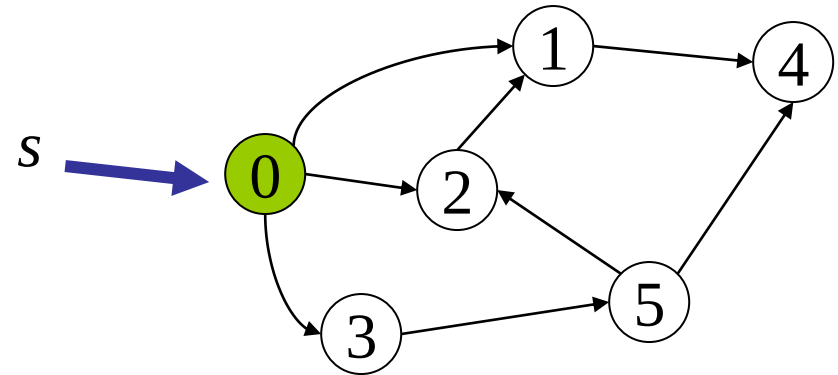
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$ 0

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

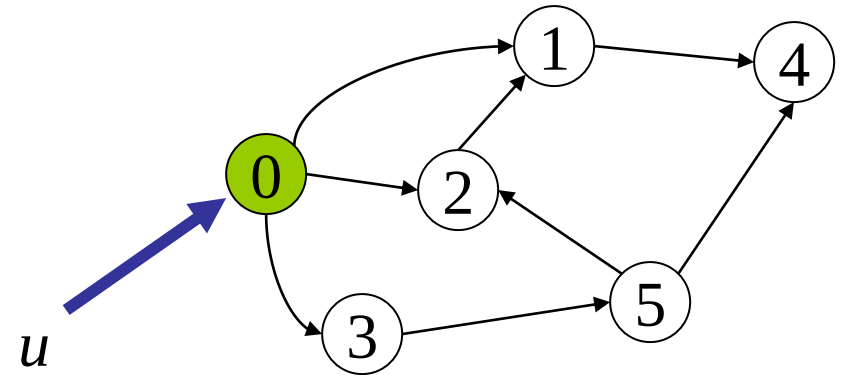
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$ 0

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

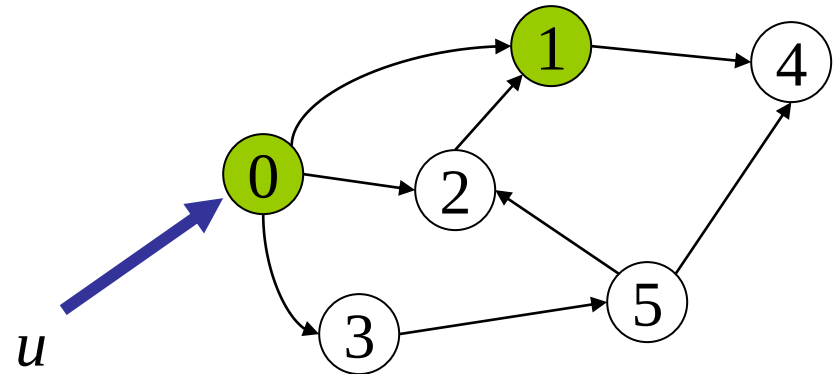
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

0	1
---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

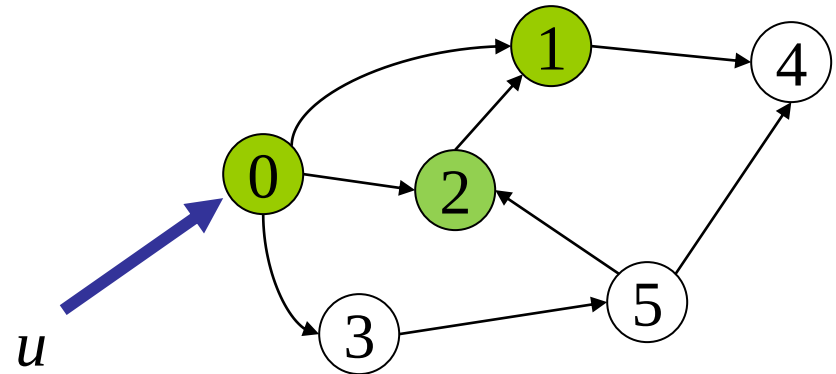
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

0	1	2
---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

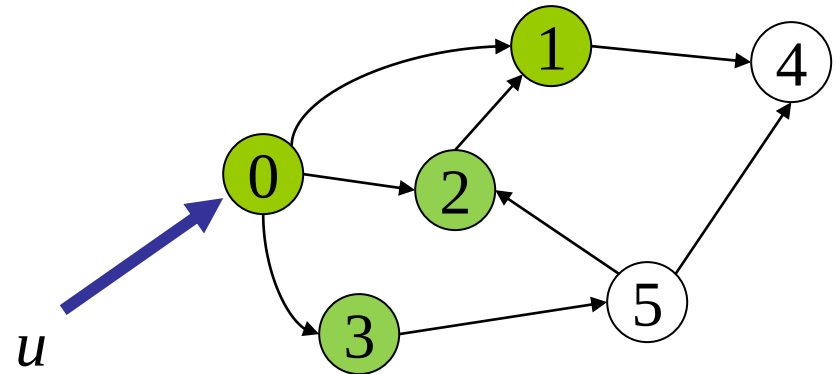
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

0	1	2	3
---	---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

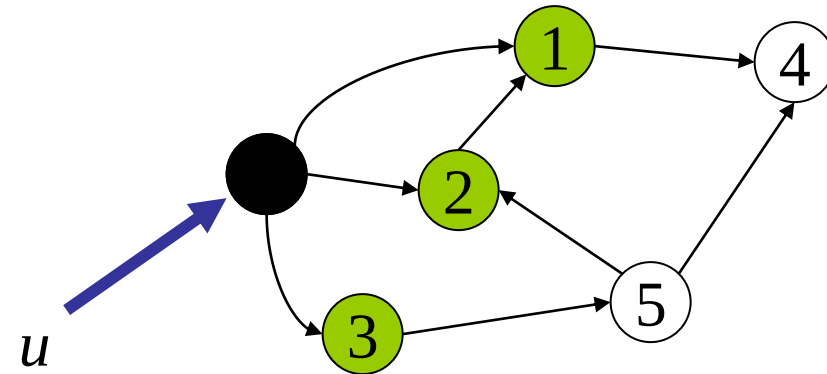
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

1	2	3
---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

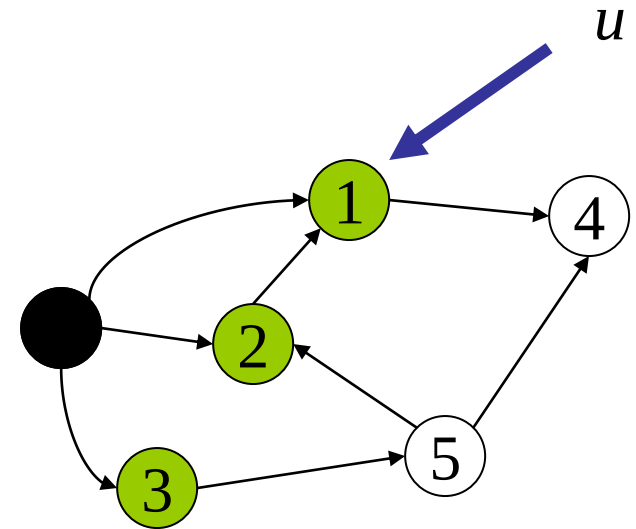
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

1	2	3
---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

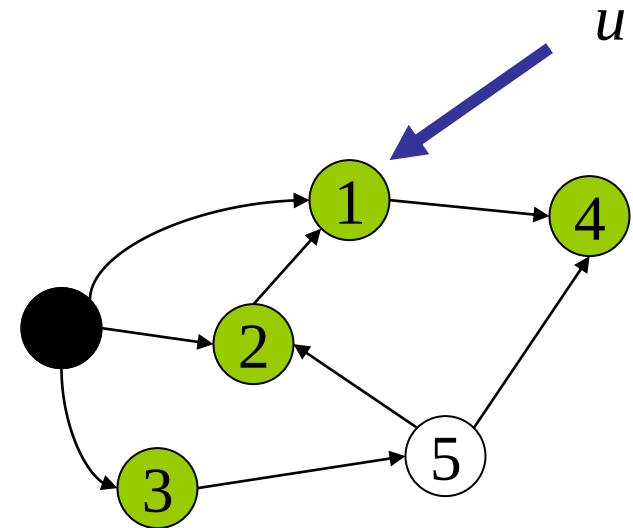
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

1	2	3	4
---	---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex
 $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

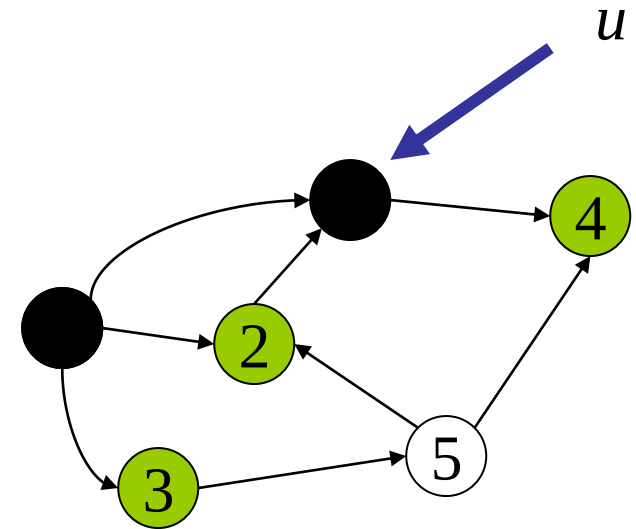
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

2	3	4
---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

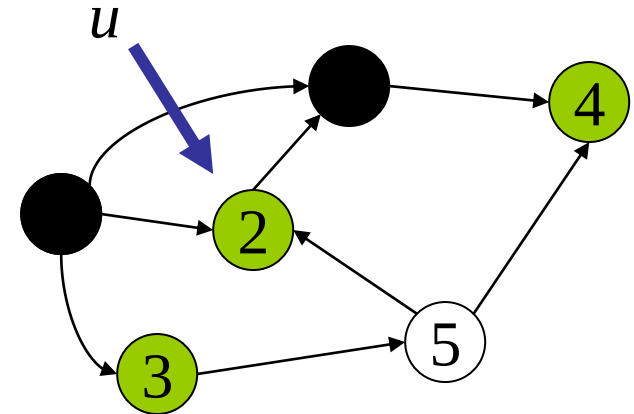
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

2	3	4
---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex s
 $\in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

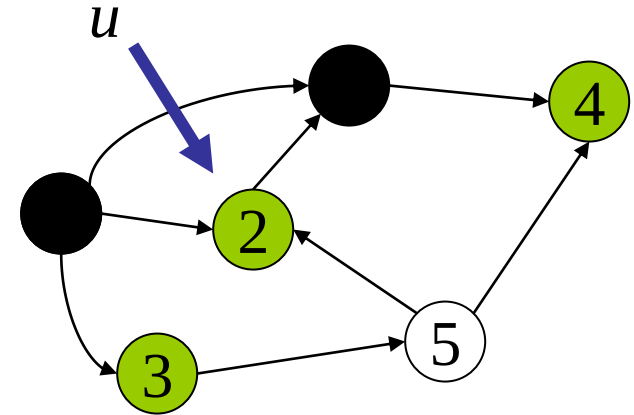
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

2	3	4
---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

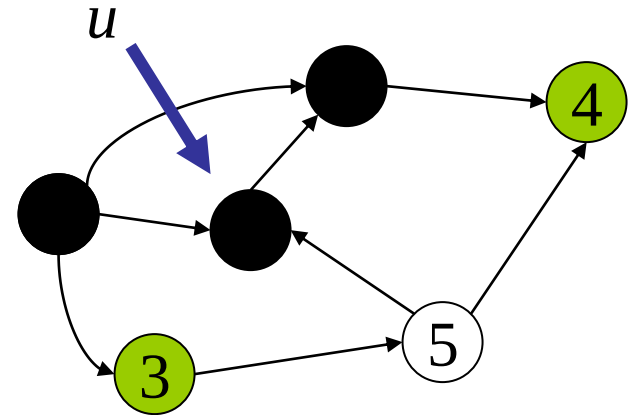
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

3	4
---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex s
 $\in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

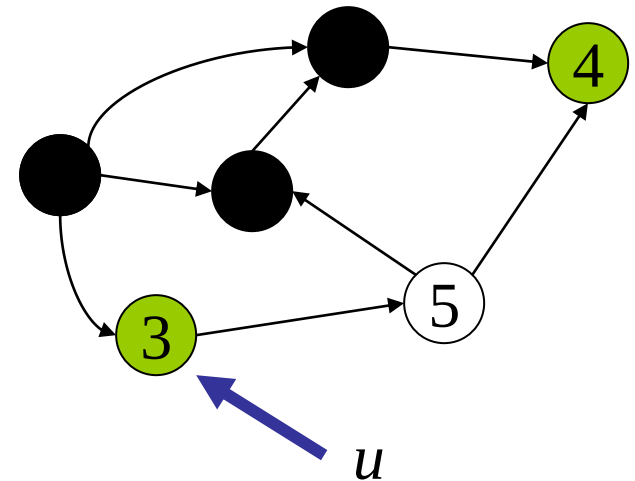
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

3	4
---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

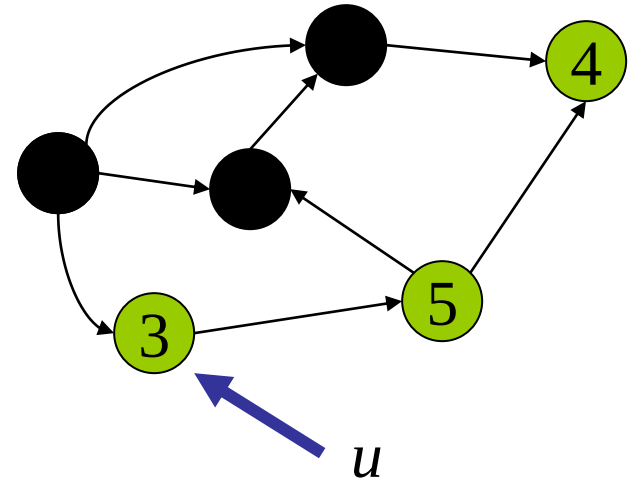
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

3	4	5
---	---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex s
 $\in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

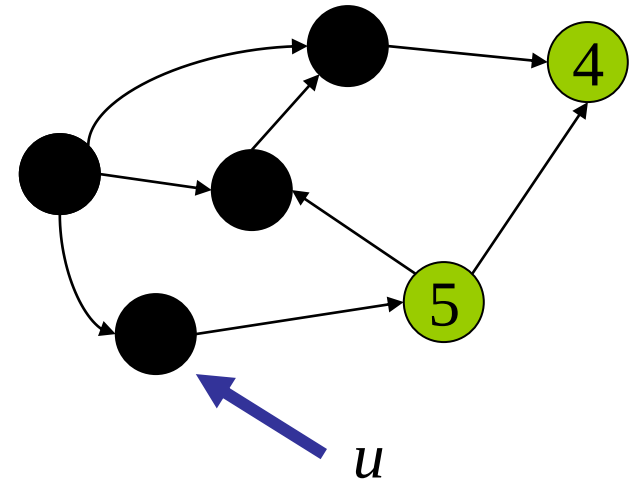
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

4	5
---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex
 $s \in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

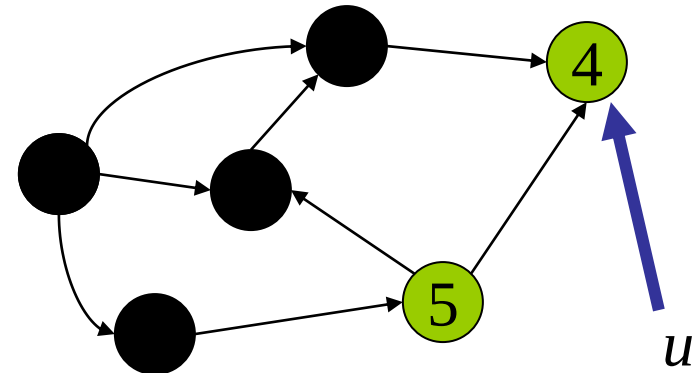
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$

4	5
---	---

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex s
 $\in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

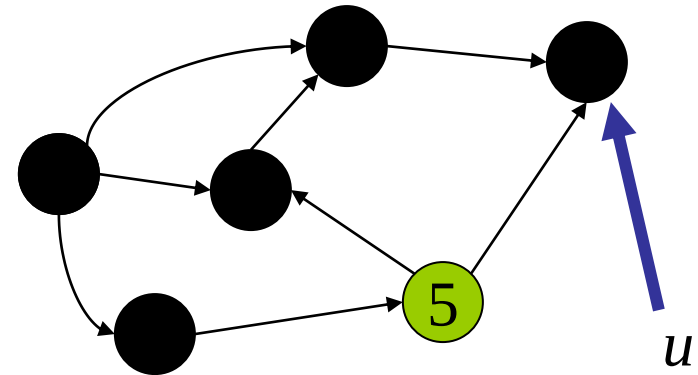
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



$Q =$ 5

Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex s
 $\in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

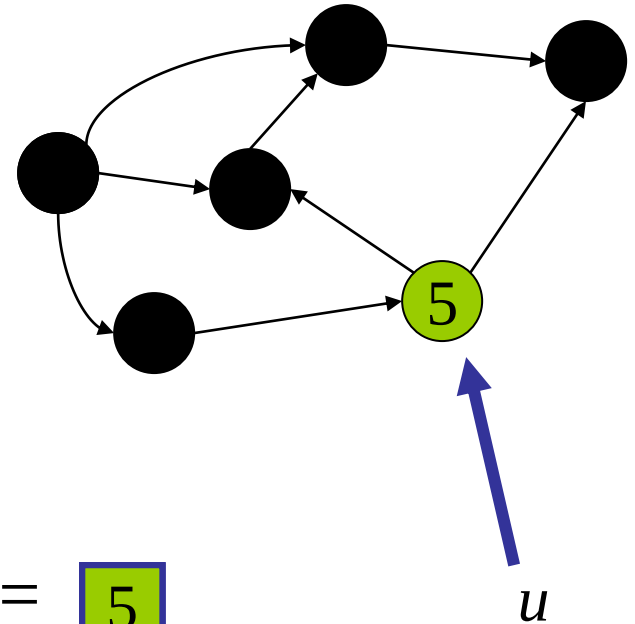
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



Graph Traversal: BFS

Given graph $G=(V,E)$ and source vertex s
 $\in V$

Create a queue Q

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$Q \leftarrow \{s\}$

While $Q \neq \emptyset$

{

$u \leftarrow head[Q];$

for each $v \in Adjacent[u]$

if $color[v] = white$

{

$color[v] \leftarrow green$

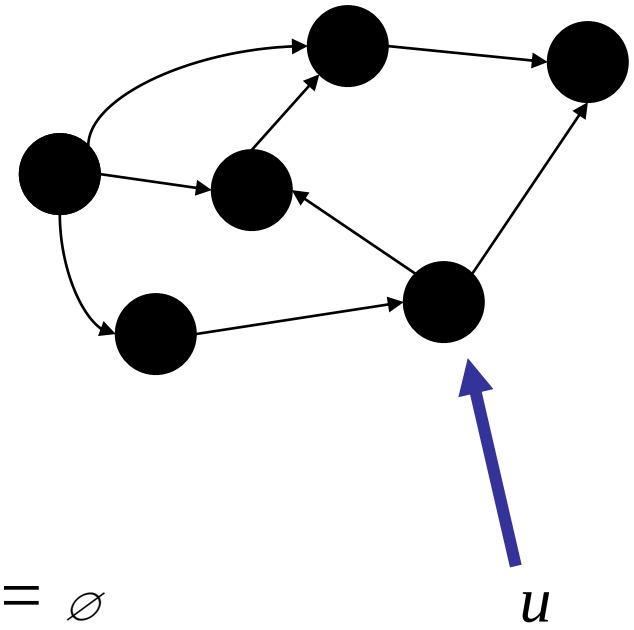
Enqueue(Q, v)

}

Dequeue(Q)

$color[u] \leftarrow black;$

}



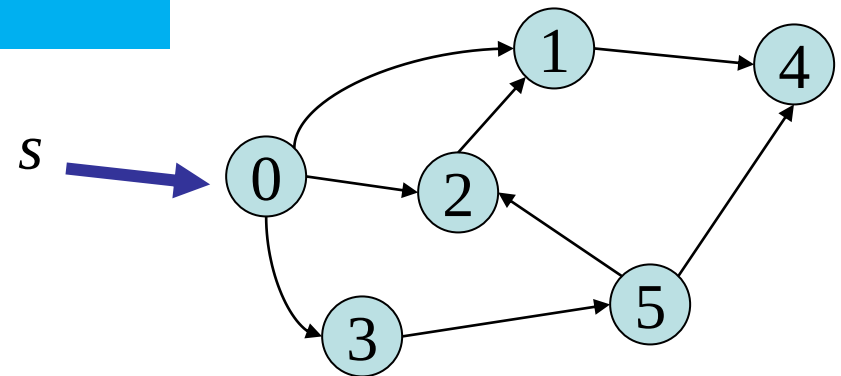
BFS Analysis

- The *while-loop* in breadth-first search is executed *at most $|V|$ times*. The reason is that every vertex enqueued at most once. So, we have $O(V)$.
- The for-loop inside the while-loop is executed *at most $|E|$ times* if G is a directed graph or *$2|E|$ times* if G is undirected.
- Therefore, the total running time for breadth-first search traversal is $O(V + E)$.

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$
Create a stack S

```
For each vertex  $u \in V - \{s\}$   
     $color[u] \leftarrow white$   
 $color[s] \leftarrow green$   
 $S \leftarrow \{s\}$   
While  $S \neq \emptyset$   
{  
     $u = Pop(S)$   
    for each  $v \in Adjacent[u]$   
        if  $color[v] = white$   
        {  
             $color[v] \leftarrow green$   
            Push( $S, v$ )  
        }  
     $color[u] \leftarrow black;$   
}
```



$S = \emptyset$

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$
 $color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

 for each $v \in \text{Adjacent}[u]$

 if $color[v] = white$

 {

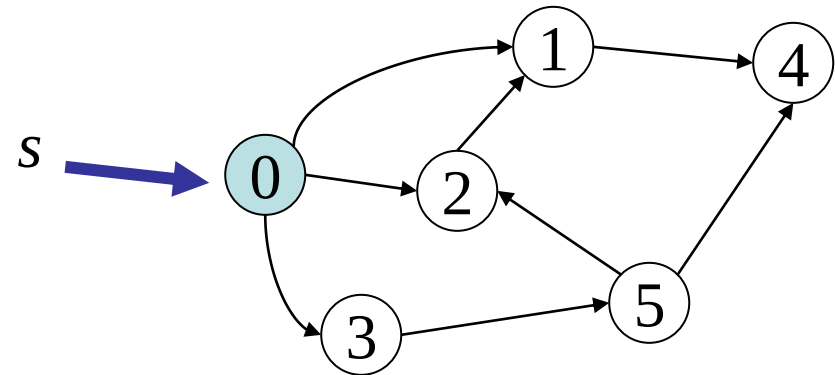
$color[v] \leftarrow green$

$\text{Push}(S,v)$

 }

$color[u] \leftarrow black;$

}



$S = \emptyset$

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

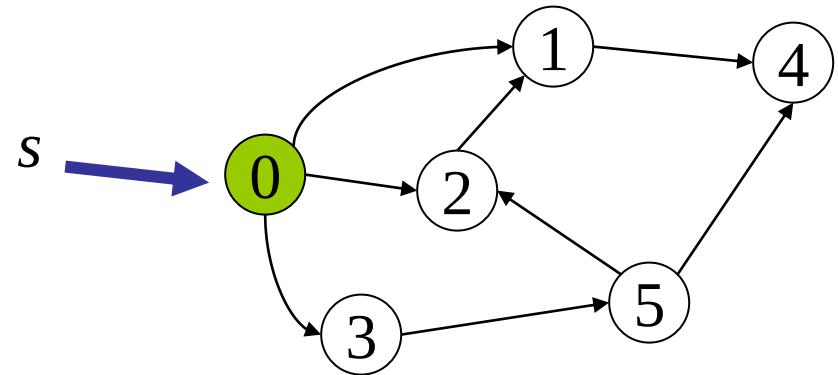
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$ 0

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

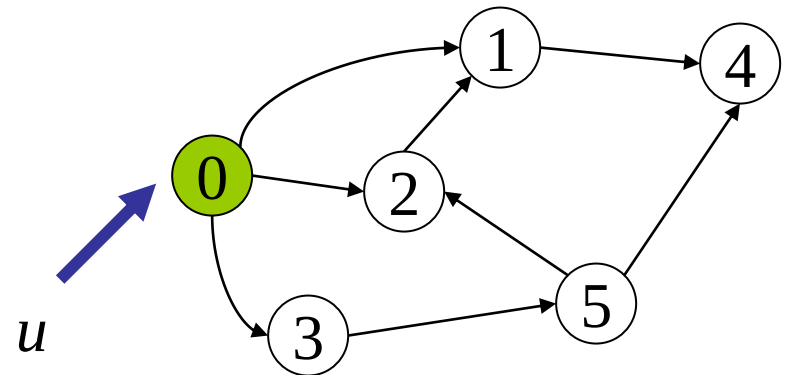
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S = \emptyset$

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

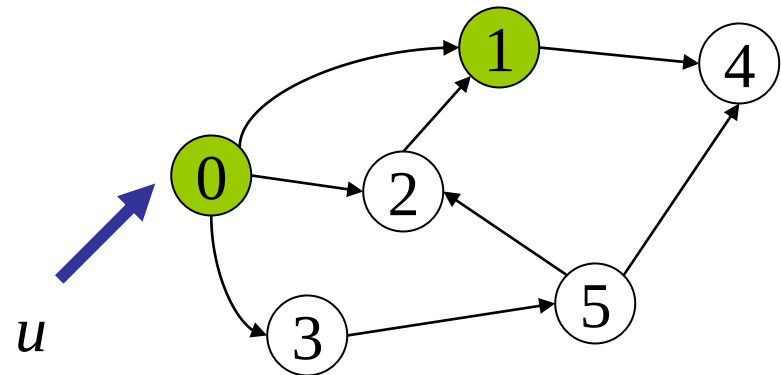
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$ 1

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

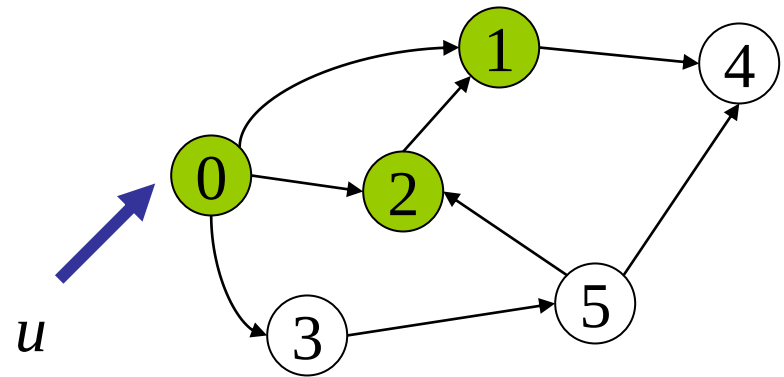
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$

2	1
---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

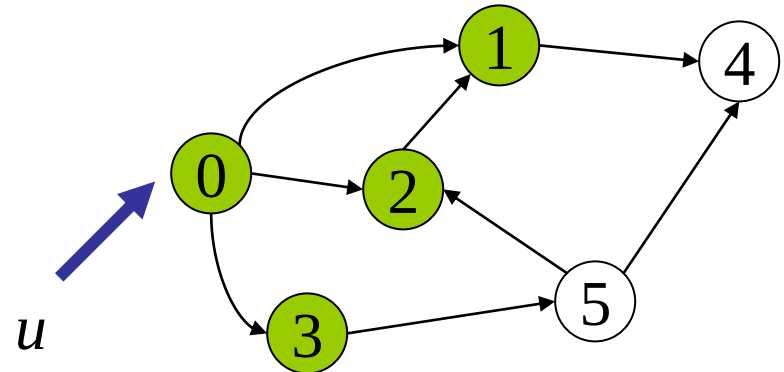
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$

3	2	1
---	---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

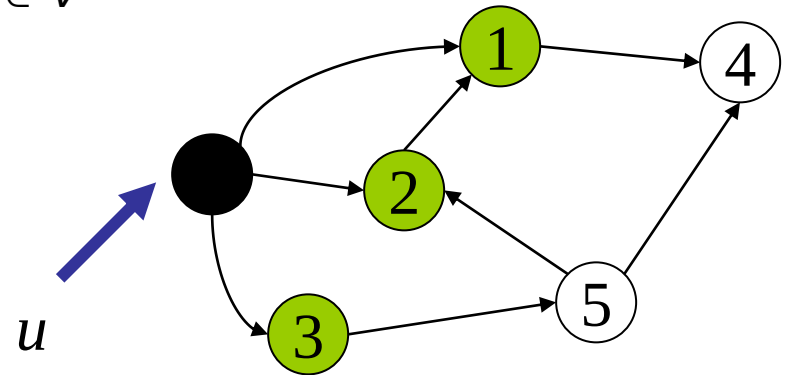
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$

3	2	1
---	---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

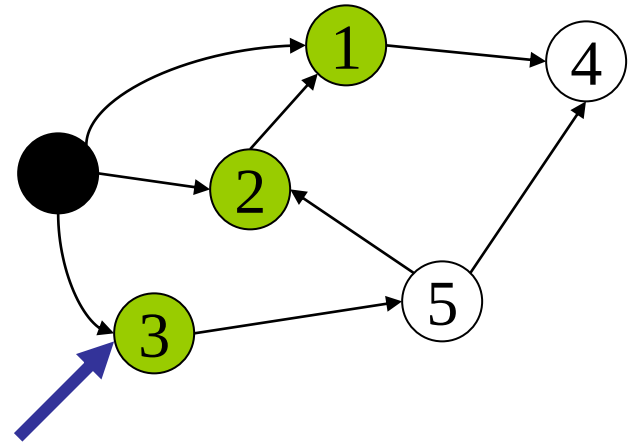
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



u

$S =$

2	1
---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

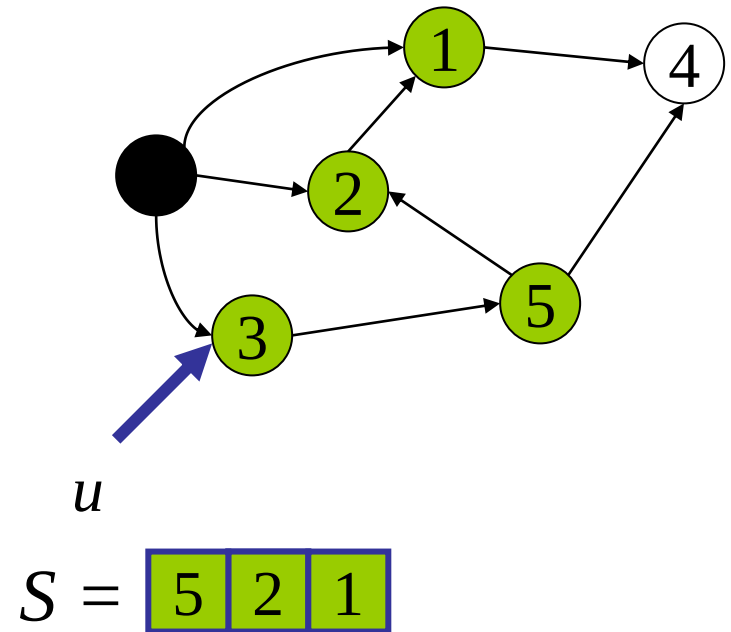
$color[v] \leftarrow green$

$\text{Push}(S, v)$

}

$color[u] \leftarrow black;$

}



Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

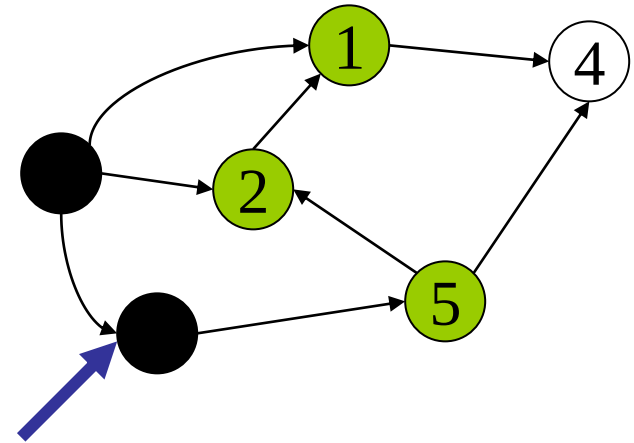
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



u

$S =$

5	2	1
---	---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

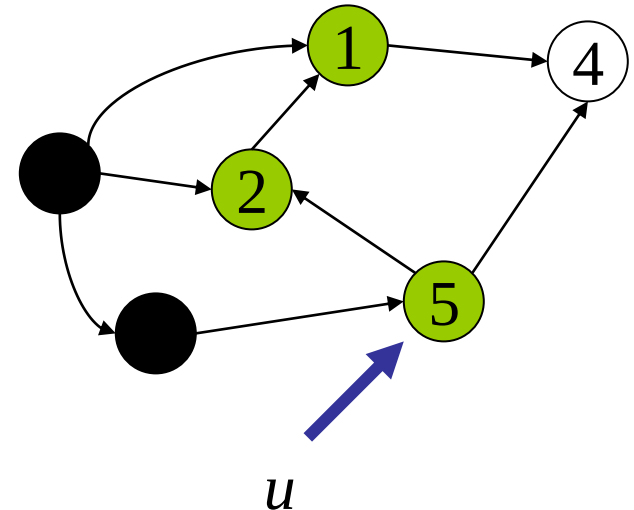
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$

2	1
---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

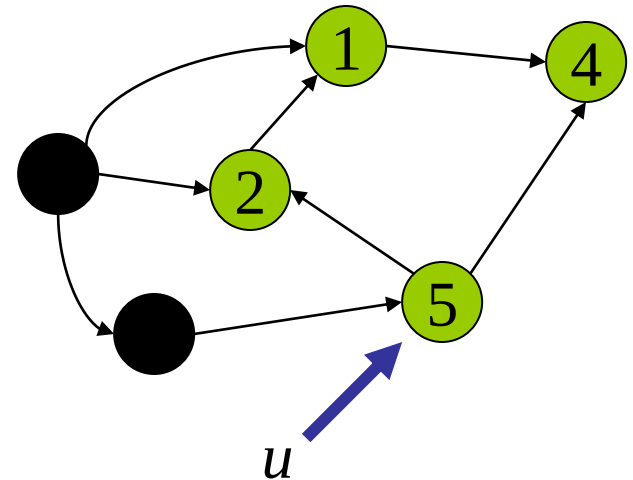
$color[v] \leftarrow green$

$\text{Push}(S,v)$

}

$color[u] \leftarrow black;$

}



$S =$

4	2	1
---	---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

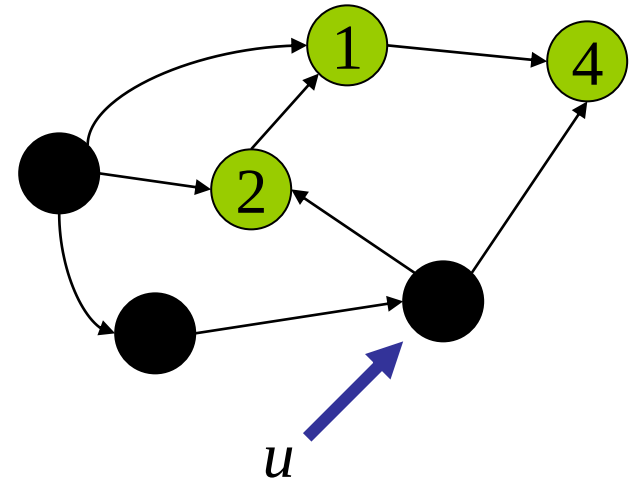
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$

4	2	1
---	---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

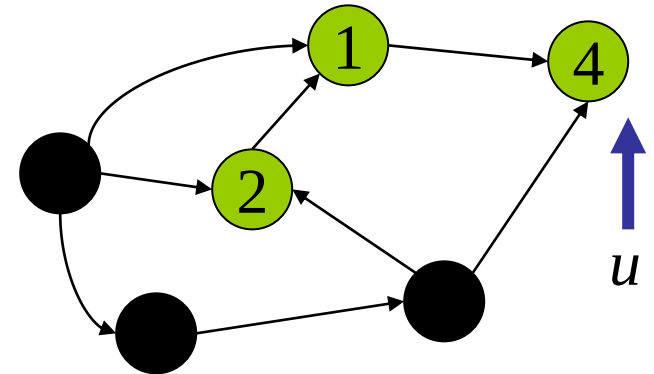
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$

2	1
---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

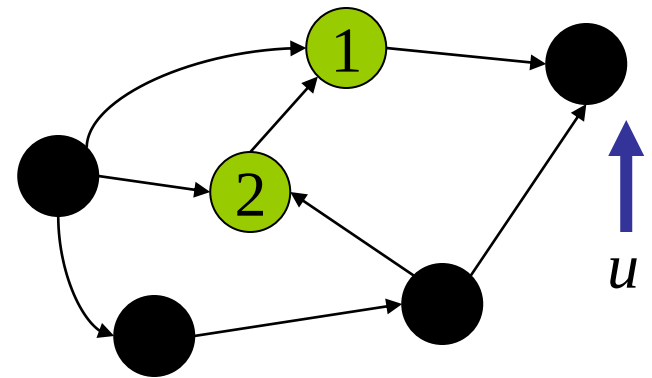
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$

2	1
---	---

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

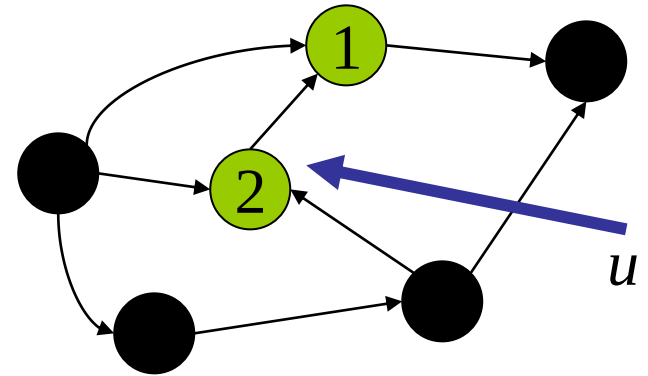
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S =$ 1

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

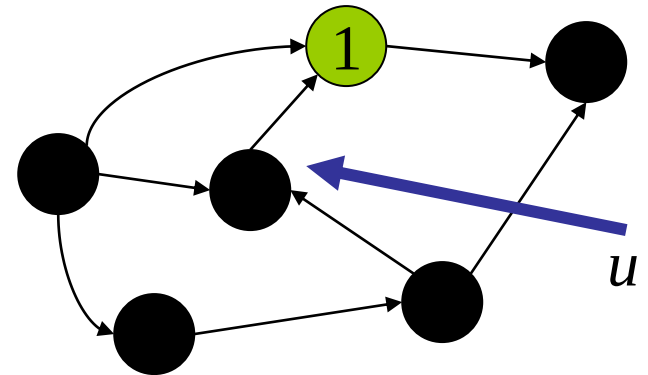
$color[v] \leftarrow green$

$\text{Push}(S,v)$

}

$color[u] \leftarrow black;$

}



$S =$ 1

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

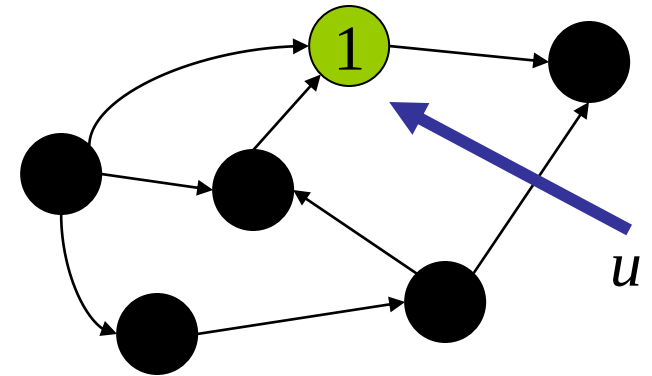
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S = \emptyset$

Graph Traversal: DFS

Given graph $G=(V,E)$ and source vertex $s \in V$

Create a stack S

For each vertex $u \in V - \{s\}$

$color[u] \leftarrow white$

$color[s] \leftarrow green$

$S \leftarrow \{s\}$

While $S \neq \emptyset$

{

$u = \text{Pop}(S)$

for each $v \in \text{Adjacent}[u]$

if $color[v] = white$

{

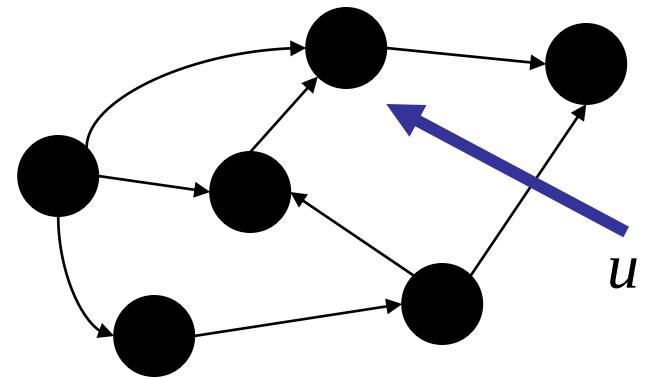
$color[v] \leftarrow green$

Push(S, v)

}

$color[u] \leftarrow black;$

}



$S = \emptyset$